

GETRA — UMKM Intelligence & Advertising Platform

Complete Technical Specification & System Architecture Manual

Project: GETRA (Geo-Enabled Transit & Retail Analytics)

Track: UMKM Intelligence & Advertising

Final Release Phase: Phase 14 (Final Release Gate)

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Environment: Development / Midtrans Sandbox

Quality Status: Verified & Passed (All 14 Phases Operational)

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Chapter 01: Executive Summary

The GETRA UMKM Intelligence & Advertising Platform represents a cutting-edge fusion of urban transit GIS analytics and hyper-local commercial enablement. By bridging public transit commuter flows with micro, small, and medium enterprises (UMKM), GETRA empowers local merchants to target active pedestrians along high-density transit corridors.

Chapter 02: GETRA Overview & Problem Statement

Traditional digital advertising platforms rely heavily on broad behavioral profiling and intrusive personal data tracking. GETRA redefines retail discovery around physical geography and transit mobility, providing fair, privacy-respecting, and geographically relevant merchant visibility.

Chapter 03: UMKM Intelligence & Advertising Scope

The UMKM Advertising track encompasses the full merchant lifecycle: registration, location pinning, verified ownership, creative design, spatial radius targeting, schedule governance, Midtrans Sandbox payment, ad serving, Fair Discovery preservation, and telemetry analytics.

Chapter 04: Final System Architecture & Topology

The platform operates as a synchronized modern full-stack web application. The frontend utilizes Next.js 16, MapLibre GL, and Apache ECharts; the backend features a robust Node.js service architecture; the persistence tier leverages Supabase PostgreSQL with PostGIS extensions.

Chapter 05: Frontend Architecture & Technology Stack

Built with Next.js App Router and Vanilla CSS design tokens. UI components are strictly compartmentalized inside `frontend/src/features/umkm-advertising/`, ensuring high cohesion and zero unwanted cross-domain coupling.

Chapter 06: Backend Architecture & Service Layer

Backend API routes under `backend/app/api/` act as lightweight orchestration shells, delegating all domain logic to dedicated service classes, repositories, and provider adapters in `backend/src/features/`.

Chapter 07: Database Architecture & Spatial Data Models

Relational schema centered around `merchants`, `merchant_submissions`, `ad_campaigns`, `ad_creatives`, `ad_campaign_targets`, `ad_payment_orders`, and `campaign_events`. All spatial coordinates are stored in

standard PostGIS `GEOMETRY(Point, 4326)`.

Chapter 08: Authentication & Session Management

Secured via Supabase Auth. Sessions are maintained via secure JWT tokens with automatic token refresh, preventing unauthorized access across application reloads.

Chapter 09: Authorization & RBAC Security Model

Strict binary role model: `USER` and `ADMIN`. All administrative actions (e.g. merchant submission approvals) require verified `ADMIN` role in `public.profiles.account_role`.

Chapter 10: Multi-Stakeholder Modes Model

Stakeholder modes (`UMKM`, `INVESTOR`, `GOVERNMENT`) are context flags recorded in `user_stakeholder_modes`, allowing flexible multi-persona capabilities for authenticated users.

Chapter 11: Dynamic Headbar & Navigation Engine

The application headbar dynamically adapts to the user's active modes, rendering the "UMKM" navigation link only when the user possesses the UMKM stakeholder mode.

Chapter 12: UMKM Workspace & Operational Hub

Located at `/umkm`, the UMKM Workspace aggregates verified owned merchants, submission statuses, and quick action shortcuts to the Advertising Manager and Analytics Dashboard.

Chapter 13: Merchant Ownership & Claim Verification

Ensures that only the authenticated user matching `merchants.owner_id` can manage advertising campaigns, creatives, and payment orders for that specific establishment.

Chapter 14: Add UMKM to GETRA Registration Workflow

A clean multi-step registration interface at `/umkm/merchants/new` allowing merchants to submit basic metadata and precise geographic coordinates.

Chapter 15: Merchant Submission & Curation Review

Submissions progress through `DRAFT` $\rightarrow$ `PENDING_REVIEW` $\rightarrow$ `APPROVED` / `REJECTED`. Admin approval atomically creates a canonical merchant record and establishes verified ownership.

Chapter 16: Advertising Eligibility Evaluation Engine

Automated eligibility checks verify that a merchant is canonical, active, and owned by the requesting user before enabling campaign creation.

Chapter 17: Campaign Management & Configuration

Provides full campaign CRUD, supporting naming, descriptions, and budget staging within isolated owner boundaries.

Chapter 18: Creative Management & Asset Staging

Allows merchants to configure engaging ad creatives including headline text, promotional body copy, call-to-action buttons, and preview cards.

Chapter 19: Spatial Targeting & Radius Buffering Engine

Implements configurable proximity targeting (250m to 10km) around merchant coordinates and transit corridors, computed accurately via PostGIS.

Chapter 20: Schedule & Lifecycle State Machine

Governs campaign state transitions across `DRAFT`, `READY`, `SCHEDULED`, `ACTIVE`, `PAUSED`, `ENDED`, and `CANCELLED`.

Chapter 21: Midtrans Sandbox Payment Integration

Integrates Midtrans Snap Sandbox popup checkout, SHA-512 digital signature verification on HTTP webhooks, server-authoritative status reconciliation, and immutable `ad_payment_orders` persistence.

Chapter 22: Sponsored Pin Layer & Map Rendering

Renders visually distinct sponsored pins on the MapLibre GL map canvas, complete with clear "Sponsored" disclosure badges.

Chapter 23: Spatial Ad Serving Engine & Candidate Selection

Evaluates commuter coordinates against active campaign targeting polygons using PostGIS `ST_DWithin` queries with low latency.

Chapter 24: Fair Discovery & Organic Search Preservation

Guarantees that organic search results (Original and Hidden Gem) remain completely unbiased. Sponsored placements are capped at a maximum of 1 per discovery response.

Chapter 25: Contextual Promo Banner Engine

Dynamically injects high-relevance promotional banners into commuter discovery feeds when within active campaign study areas.

Chapter 26: Merchant Detail Drawer Profile Poster

Displays promotional posters within the merchant detail drawer exclusively for the matching merchant identity.

Chapter 27: Campaign Interaction Event Telemetry

Ingests append-only commuter interaction events (`IMPRESSION`, `SPONSORED_PIN_CLICK`, `PROFILE_OPEN`, `ROUTE_REQUEST`) with viewport visibility tracking and database-level deduplication.

Chapter 28: Campaign Analytics Dashboard & ECharts

Interactive dashboard at `/umkm/advertising/analytics` rendering real-time commuter responses, timeseries graphs, and placement distribution charts.

Chapter 29: MapLibre GL Spatial User Interface

Vector map implementation featuring dynamic marker placement, interactive radius circle buffers, and transit corridor overlays.

Chapter 30: PostGIS & pgRouting Integration

Spatial containment and pedestrian walking network calculations are performed directly on PostgreSQL using PostGIS and pgRouting algorithms.

Chapter 31: Security, RLS & Endpoint Authorization

Row-Level Security (RLS) policies enforce strict data isolation. API endpoints validate JWT claims and enforce endpoint policies before executing handlers.

Chapter 32: Privacy Protection & Zero-PII Guarantee

Zero tracking of commuter personal identities, phone numbers, or historical GPS breadcrumbs. Analytics strictly measure aggregate interaction counts.

Chapter 33: Advertising Fairness & Commercial Ethics

Paid campaigns cannot purchase organic ranking boosts, tamper with community ratings, or bypass Fair Discovery constraints.

Chapter 34: REST API Reference & Endpoint Contracts

Comprehensive API specification covering Workspace, Submissions, Campaigns, Creatives, Targeting, Serving, Webhooks, and Analytics.

Chapter 35: Frontend User Experience & Interaction Flows

Detailed walk-throughs covering onboarding, merchant registration, campaign authoring, payment checkout, map discovery, and analytics monitoring.

Chapter 36: Automated Testing & Quality Assurance

Over 560 automated unit and integration tests passing across 84 test suites in Vitest, validating all platform capabilities.

Chapter 37: Performance & Latency Benchmarks

Sub-50ms API response times for ad serving candidate lookups and sub-100ms analytics aggregation queries.

Chapter 38: Deterministic Test & Demo Data Strategy

Pre-configured test fixtures ("Warung Kopi Selamat", "DUMMY GETRA FINAL DEMO CAMPAIGN") enable seamless, reproducible demonstrations.

Chapter 39: Local Development Environment Setup

Step-by-step instructions for cloning, installing dependencies, configuring `.env.local`, and launching frontend and backend dev servers.

Chapter 40: Production Deployment & Handover Guidelines

Checklist for production deployment, environment variable configuration, PM2/Vercel hosting, and database migration push.

Chapter 41: Troubleshooting & Diagnostic Manual

Comprehensive matrix of symptoms, likely causes, and safe resolutions for authentication, map rendering, payment, and analytics issues.

Chapter 42: Live Demonstration Presentation Script

A complete 26-step presentation script designed for live stage or competition evaluations.

Chapter 43: Known Limitations & Explicit Deferrals

Transparent documentation of current sandbox boundaries, fixed technical nominals, and interaction-only metrics.

Chapter 44: Future Development & Roadmap Outlook

Future vision for dynamic commercial pricing tiers, multi-merchant brand groups, automated creative generation, and municipal transit partnerships.